1. Jackson, Problem 5.1

Consider the i-th cartesian component of the B-Field,

$$
\begin{aligned}
\frac{4\pi}{\mu_0 I}\mathbf{B}(\mathbf{x})\cdot\hat{\mathbf{x}}_i &= \oint_{\partial S}\hat{\mathbf{x}}_i\cdot\left[d\mathbf{l}'\times\frac{(\mathbf{x}-\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|}\right] \\
&= \oint_{\partial S}\hat{\mathbf{x}}_i\cdot\left[d\mathbf{l}'\times\nabla_{x'}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right] \\
&= \oint_{\partial S}d\mathbf{l}'\cdot\left[\nabla_{x'}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\times\hat{\mathbf{x}}_i\right] \quad | \quad \text{Stokes' law} \\
&= \int_S d\mathbf{a}'\cdot\left\{\nabla_{x'}\times\left[\nabla_{x'}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\times\hat{\mathbf{x}}_i\right]\right\} \\
&= \int_S d\mathbf{a}'\cdot\left\{\left[\nabla_{x'}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right](\nabla_{x'}\cdot\hat{\mathbf{x}}_i)-\hat{\mathbf{x}}_i\left[\nabla^2_{x'}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right]\right. \\
&\quad \left.+(\hat{\mathbf{x}}_i\cdot\nabla_{x'})\left[\nabla_{x'}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right]-(\left[\nabla_{x'}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right]\cdot\nabla_{x'})\hat{\mathbf{x}}_i\right\} \quad \text{use} \quad \mathbf{x}\neq\mathbf{x}' \quad \text{always} \\
&= \int_S d\mathbf{a}'\cdot\left\{0-0+(\hat{\mathbf{x}}_i\cdot\nabla_{x'})\left[\nabla_{x'}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right]-0\right\} \\
&= \int_S d\mathbf{a}'\cdot\left\{0-0+(\hat{\mathbf{x}}_i\cdot\nabla_{x'})\left[\nabla_{x'}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right]-0\right\} \\
&= \int_S d\mathbf{a}'\cdot\left\{\frac{\partial}{\partial x_i'}\left[\nabla_{x'}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right]\right\} = -\int_S d\mathbf{a}'\cdot\left\{\frac{\partial}{\partial x_i}\left[\nabla_{x'}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right]\right\} \\
&= -\frac{\partial}{\partial x_i}\int_S d\mathbf{a}'\cdot\left\{\left[\nabla_{x'}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right]\right\} \quad | \quad \text{see Eqn. after 1.25 on page 33 of textbook} \\
&= \frac{\partial}{\partial x_i}\int_S d\Omega' = \frac{\partial}{\partial x_i}\Omega(\mathbf{x})
\end{aligned}
$$

Thus, $B_i = \frac{\mu_0 I}{4\pi}\frac{\partial}{\partial x_i}\Omega(\mathbf{x})$, and

$$\mathbf{B}(\mathbf{x}) = \frac{\mu_0 I}{4\pi}\nabla_x\Omega(\mathbf{x}) \quad \text{q.e.d.}$$

2. Jackson, Problem 5.3

Consider a loop current with radius R around the z-axis. The loop is centered at location $\hat{\mathbf{z}}z'$. Then, the magnetic field at an observation point $\hat{\mathbf{z}}z$ is

$$\mathbf{B}(0,0,z) = \frac{\mu_0 I}{4\pi} \oint d\mathbf{l}' \times \frac{\mathbf{x}-\mathbf{x}'}{|\mathbf{x}-\mathbf{x}'|^3}$$

Insert $\mathbf{x}' = \begin{pmatrix} R\cos\phi' \\ R\sin\phi' \\ z' \end{pmatrix}$ and $d\mathbf{l}' = \begin{pmatrix} -R\sin\phi' \\ R\cos\phi' \\ 0 \end{pmatrix} d\phi'$ to find

$$\mathbf{B}(0,0,z) = \frac{\mu_0 I}{4\pi} \frac{1}{\sqrt{R^2+z'^2}^3} \int_0^{2\pi} \begin{pmatrix} -Rz'\cos\phi' \\ Rz'\sin\phi' \\ R^2 \end{pmatrix} d\phi' = \hat{\mathbf{z}} \frac{\mu_0 I}{2} \frac{R^2}{\sqrt{R^2+z'^2}^3}$$

Now, consider a solenoid the axis of which coincides with the z-axis. The solenoid has N windings per length, current I, and end points z_1 and z_2. Then, the amount of current flowing across a length dz' is $dI = INdz'$, and

$$\begin{aligned}
\mathbf{B}(0,0,z) &= \int \hat{\mathbf{z}} \frac{\mu_0 dI}{2} \frac{R^2}{\sqrt{R^2+z'^2}^3} \\
&= \hat{\mathbf{z}} \frac{\mu_0 R^2 IN}{2} \int_{z_1}^{z_2} \frac{dz'}{\sqrt{R^2+z'^2}^3} \\
&= \hat{\mathbf{z}} \frac{\mu_0 R^2 IN}{2} \left[\frac{z'}{R^2\sqrt{R^2+z'^2}^3} \right]_{z_1}^{z_2} \\
&= \hat{\mathbf{z}} \frac{\mu_0 IN}{2} \left[\cos\theta_1 + \cos\theta_2\right] \quad ,
\end{aligned}$$

with angles θ_1 and θ_2 as shown in the problem statement.

5.4 A magnetic induction $\vec{B}$ in a current-free region in a uniform medium is cylindrically symmetric with components $B_z(\rho, z)$ and $B_\rho(\rho, z)$ and with a known $B_z(0, z)$ on the axis of symmetry. The magnitude of the axial field varies slowly in z.

a) Show that near the axis the axial and radial components of magnetic induction are approximately

$$B_z(\rho, z) \approx B_z(0, z) - \left(\frac{\rho^2}{4}\right)\left[\frac{\partial^2 B_z(0, z)}{\partial z^2}\right] + \cdots$$
$$B_\rho(\rho, z) \approx -\left(\frac{\rho}{2}\right)\left[\frac{\partial B_z(0, z)}{\partial z}\right] + \left(\frac{\rho^3}{16}\right)\left[\frac{\partial^3 B_z(0, z)}{\partial z^3}\right] + \cdots$$

Near the axis ($\rho \approx 0$), we may perform a series expansion of the magnetic induction

$$\begin{aligned} B_z(\rho, z) &= b_0(z) + \rho b_1(z) + \tfrac{1}{2}\rho^2 b_2(z) + \cdots = \sum_{n=0}^{\infty} \frac{\rho^n}{n!} b_n(z) \\ B_\rho(\rho, z) &= c_0(z) + \rho c_1(z) + \tfrac{1}{2}\rho^2 c_2(z) + \cdots = \sum_{n=0}^{\infty} \frac{\rho^n}{n!} c_n(z) \\ B_\phi(\rho, z) &= 0 \end{aligned} \tag{1}$$

Since the region is current free, the magnetic induction must satisfy the magnetostatic equations $\vec{\nabla} \cdot \vec{B} = 0$ and $\vec{\nabla} \times \vec{B} = 0$. In cylindrical coordinates, the divergence equation gives

$$0 = \vec{\nabla} \cdot \vec{B} = \frac{1}{\rho}\frac{\partial}{\partial \rho}\rho B_\rho + \frac{\partial}{\partial z} B_z = \sum_{n=0}^{\infty}\left(\frac{(n+1)\rho^{n-1}}{n!} c_n(z) + \frac{\rho^n}{n!} b_n'(z)\right)$$

We shift $n \to n+1$ in the first term in the sum to obtain

$$0 = \frac{1}{\rho} c_0(z) + \sum_{n=0}^{\infty} \frac{\rho^n}{n!}\left(\frac{n+2}{n+1} c_{n+1}(z) + b_n'(z)\right)$$

Since this vanishes for any value of ρ, each term in the sum must vanish individually. This gives us

$$c_0(z) = 0, \qquad c_{n+1} = -\frac{n+1}{n+2} b_n'(z) \tag{2}$$

We now continue with the curl equation. The only non-trivial component is

$$0 = [\vec{\nabla} \times \vec{B}]_\phi = \frac{\partial}{\partial z} B_\rho - \frac{\partial}{\partial \rho} B_z = \sum_{n=0}^{\infty} \left(\frac{\rho^n}{n!} c_n'(z) - \frac{\rho^{n-1}}{(n-1)!} b_n(z) \right)$$

Shifting $n \to n+1$ in the second term in the sum gives

$$0 = \sum_{n=0}^{\infty} \frac{\rho^n}{n!} \left(c_n'(z) - b_{n+1}(z) \right)$$

so that

$$b_{n+1}(z) = c_n'(z)$$

Combining this with (2) gives us the recursion relation

$$b_{n+1}(z) = c_n'(z) = -\frac{n}{n+1} b_{n-1}''(z)$$

Taking into account the fact that $c_0(z) = 0$, we see that $b_1(z) = 0$. On the other hand, $b_0(z)$ is undetermined. The recursion relation can then be used to relate $b_n(z)$ to $b_0(z)$ where n is an even integer. The result is

$$\begin{aligned} b_n(z) &= (-1)^{n/2} \frac{(n-1)(n-3)\cdots 3 \cdot 1}{n(n-2)\cdots 4 \cdot 2} b_0^{(n)}(z) \\ &= \frac{(-1)^{n/2}}{2^n} \frac{n!}{[(n/2)!]^2} b_0^{(n)}(z) \qquad (n \text{ even}) \end{aligned} \tag{3}$$

where $b_0^{(n)}(z)$ indicates the n-th derivative of $b_0(z)$. Using (2), we then see that the odd $c_n(z)$ terms are non-vanishing, and are given by

$$c_{n+1}(z) = \frac{(-1)^{n/2+1}}{2^n} \frac{(n+1)!}{(n+2)[(n/2)!]^2} b_0^{(n+1)}(z) \qquad (n \text{ odd})$$

Inserting these expressions into the Taylor expansion (1) finally gives

$$\begin{aligned} B_z(\rho, z) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{4^k} \frac{\rho^{2k}}{(k!)^2} \left[\frac{\partial^{2k} B_z(0,z)}{\partial z^{2k}} \right] \\ B_\rho(\rho, z) &= \sum_{k=0}^{\infty} \frac{(-1)^{k+1}}{4^k} \frac{\rho^{(2k+1)}}{(2k+2)(k!)^2} \left[\frac{\partial^{2k+1} B_z(0,z)}{\partial z^{2k+1}} \right] \end{aligned} \tag{4}$$

where we have made the identification $b_0(z) = B_z(0, z)$. The first few terms in the expansion gives the desired result

$$\begin{aligned} B_z(\rho, z) &= B_z(0,z) - \frac{\rho^2}{4} \left[\frac{\partial^2 B_z(0,z)}{\partial z^2} \right] + \cdots \\ B_\rho(\rho, z) &= -\frac{\rho}{2} \left[\frac{\partial B_z(0,z)}{\partial z} \right] + \frac{\rho^3}{16} \left[\frac{\partial^3 B_z(0,z)}{\partial z^3} \right] + \cdots \end{aligned} \tag{5}$$

Incidentally, another approach to this problem is to note that, for a current-free region, the magnetic induction may be given in terms of a magnetic scalar potential

$$\vec{B} = -\vec{\nabla}\Phi_M, \qquad \nabla^2\Phi_M = 0$$

Solving Laplace's equation in cylindrical coordinates gives an expansion in terms of Bessel functions. Choosing the modes to be oscillating in ρ and exponential in z allows us to write

$$\Phi_M(\rho, \phi, z) = \sum_m \int dk\, A_m(k) J_m(k\rho) e^{im\phi} e^{kz}$$

where $A_m(k)$ are a set of expansion coefficients. Since the problem is cylindrically symmetric, only the $m = 0$ component contributes

$$\Phi_M(\rho, z) = \int dk\, A_0(k) J_0(k\rho) e^{kz}$$

As long as we are only interested in the field near the axis, we may Taylor expand the Bessel function $J_0(k\rho)$ for small ρ. Using

$$J_n(x) = \sum_{s=0}^{\infty} \frac{(-1)^s}{s!(n+s)!} \left(\frac{x}{2}\right)^{n+2s} \tag{6}$$

we find

$$\Phi_M(\rho, z) = \sum_{s=0}^{\infty} \frac{(-1)^s}{4^s (s!)^2} \rho^{2s} \int dk\, A_0(k) k^{2s} e^{kz}$$

Since the exponential e^{kz} blows up as $z \to \infty$ (for positive k) or $z \to \infty$ (for negative k), this expression is not be well defined in all of space. However, we may consider either working in a finite range of z or treating this as a formal expression. In any case, we note that powers of k inside the integral corresponds to taking z derivatives of the exponential

$$k \quad \leftrightarrow \quad \frac{\partial}{\partial z} \tag{7}$$

Hence

$$\Phi_M(\rho, z) = \sum_{s=0}^{\infty} \frac{(-1)^s}{4^s (s!)^2} \rho^{2s} \frac{\partial^{2s}}{\partial z^{2s}} \int dk\, A_0(k) e^{kz}$$

Setting $\rho = 0$ in this expression gives

$$\Phi_M(0, z) = \int dk\, A_0(k) e^{kz}$$

Substituting this back in allows us to write

$$\Phi_M(0,z) = \sum_{s=0}^{\infty} \frac{(-1)^s}{4^s (s!)^2} \rho^{2s} \frac{\partial^{2s}}{\partial z^{2s}} \Phi_M(0,z)$$

Finally, using $\vec{B} = -\vec{\nabla}\Phi_M$ allows us to reproduce the series expansion of the magnetic induction given in (4). Incidentally, we note that the coefficients in the solution to the $b_n(z)$ recursion relation given in (3) matches the Taylor coefficients of the J_0 Bessel function, (6). In fact, if desired, we may use (7) to write the formal expression

$$\Phi_M(\rho,z) = J_0\left(\rho\frac{\partial}{\partial z}\right)\Phi_M(0,z)$$

Then

$$B_z(\rho,z) = J_0\left(\rho\frac{\partial}{\partial z}\right)B_z(0,z)$$

$$B_\rho(\rho,z) = J_0'\left(\rho\frac{\partial}{\partial z}\right)B_z(0,z) = -J_1\left(\rho\frac{\partial}{\partial z}\right)B_z(0,z)$$

b) What are the magnitudes of the neglected terms, or equivalently what is the criterion defining "near" the axis?

We see from (4) that the n-th term in the series expansion of $\vec{B}(\rho,z)$ is of the form

$$\sim \frac{\rho^n}{2^n[(n/2)!]^2}\left[\frac{\partial^n B_z(0,z)}{\partial z^n}\right]$$

Ignoring constant factors, this indicates that the ratio of adjacent terms in the series is roughly

$$\frac{a_{n+2}}{a_n} \sim \rho^2 \frac{[\partial^{n+2}B_z(0,z)/\partial z^{n+2}]}{[\partial^n B_z(0,z)/\partial z^n]}$$

In general, this ratio needs to be less than one for the series to converge. Hence, this provides a criterion for being "near" the axis

$$\rho \ll \sqrt{\frac{[\partial^n B_z(0,z)/\partial z^n]}{[\partial^{n+2}B_z(0,z)/\partial z^{n+2}]}}$$

For a smooth function $B_z(0,z)$, the n-th derivative behaves roughly as $1/L^n$ where L is the scale of variation of the field. As a result, we demand $\rho \ll L$ where L is a typical length for the variation of the magnetic induction B_z along the z direction.

5.6 A cylindrical conductor of radius a has a hole of radius b bored parallel to, and centered a distance d from, the cylinder axis ($d + b < a$). The current density is uniform throughout the remaining metal of the cylinder and is parallel to the axis. Use Ampère's law and principle of linear superposition to find the magnitude and the direction of the magnetic-flux density in the hole.

Ampère's law in integral form states

$$\oint_V \vec{B} \cdot d\vec{\ell} = \mu_0 i_{\text{enc}}$$

For a cylindrically symmetric geometry this gives simply

$$B = \frac{\mu_0 i_{\text{enc}}}{2\pi r} = \frac{\mu_0 (j\pi r^2)}{2\pi r} = \frac{\mu_0 j r}{2}$$

where we have assumed a uniform current density j. The direction of the magnetic induction is given by the right hand rule. For a conductor oriented along the z axis (so that the current is flowing in the $+\hat{z}$ direction), we may write

$$\vec{B} = \frac{\mu_0 j r}{2} \hat{z} \times \hat{r} = \frac{\mu_0 j}{2} \hat{z} \times \vec{r}$$

where $\vec{r}$ is the vector from the center of the conductor to the position where we are measuring the field. We now use linear superposition to start with a solid cylindrical conductor and then subtract the 'missing' current from the hole

$$\vec{B} = \frac{\mu_0 j}{2} \hat{z} \times \vec{x} - \frac{\mu_0 j}{2} \hat{z} \times (\vec{x} - \vec{d}) = \frac{\mu_0 j}{2} \hat{z} \times \vec{d}$$

Here $\vec{d}$ is the vector displacement of the hole from the center of the cylinder. This somewhat remarkable result demonstrates that the magnetic induction is uniform in the hole, and is in a direction given by the right hand rule.

If desired, we note that the total current carried by the wire is $I = j(\pi a^2 - \pi b^2)$, so we may express the magnetic induction in terms of I as

$$\vec{B} = \frac{\mu_0 I}{2\pi(a^2 - b^2)} \hat{z} \times \vec{d}$$